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Prediction of wildfire rate of spread in grasslands using machine learning methods

Sadegh Khanmohammadi | M. Arashpour | E. Golafshani | M. Cruz | A. Rajabifard | Yu Bai

Environmental Science, Computer Science · Environmental Modelling & Software · 1 October 2022

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Wildfire Path Prediction Spread using Machine Learning

Mapulane Makhaba | S. Winberg | Environmental Science, Computer Science ·

2022 International Conference on Electrical… · 20 July 2022

TLDR Based on the analysis of the ML models considered, the Long-term Recurrent Convolutional Network model is expected to be better as it includes spatial and temporal properties when modelling **wildfire spread**. Expand

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Multifidelity prediction in wildfire spread simulation: Modeling, uncertainty quantification and sensitivity analysis

M. M. Valero | L. Jofre | Ricardo Torres | Environmental Science, Engineering ·

Environmental Modelling & Software · 13 April 2021

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Advancing Wildfire Predictive Models: A Novel Dataset for Next-Day Wildfire Spread

Syed Haider Ali | Sunny Zhang | Saanvi Mhatre | Ting Xiao | Environmental Science ·

2024 Conference on AI, Science, Engineering, and… · 30 September 2024

Wildfire spread prediction is crucial as **wildfires** destroy millions of acres of land and produce billions of dollars in economic and insured losses annually. Recently, there has been a marked… Expand

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A Deep Learning Framework: Predicting Fire Radiative Power From the Combination of Polar-Orbiting and Geostationary Satellite Data During Wildfire Spread

Zixun Dong | Change Zheng | Fengjun Zhao | Guangyu Wang | Ye Tian | Hongchen Li

Environmental Science, Computer Science · IEEE Journal of Selected Topics in Applied Earth… · 2024

TLDR A deep learning framework is established that integrates various modules to analyze multimodal data for the accurate **prediction** of FRP imagery and has demonstrated acceptable performance in transfer learning and multistep **prediction**, emphasizing its application value in **wildfire prediction** and management. Expand

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Wildfire Prediction and Spread Detection

Ayushi Mishra | Dr. K Deeba | Environmental Science, Computer Science ·

International Conference Computing Methodologies… · 23 July 2025

TLDR This project presents a comprehensive approach to **wildfire prediction** and **spread** detection using satellite-based remote sensing data from the FIRMS (Fire Information for Resource Management System) dataset, which facilitates accurate identification of fire-prone regions, real-time fire detection, and classification of fire severity. Expand

IEEE | Save | Cite

Assimilating UAV-Based Observation Data for Wildfire Spread Simulation

Mu Ge Xiaolin Hu Environmental Science, Engineering ·

Annual Modeling and Simulation Conference · 20 May 2024

Wildfire spread simulation simulates the dynamic **spread** of **wildfires** by modeling fire **spread** behavior under various fuel, terrain, and weather conditions. To achieve real-time fire **spread prediction**… Expand

IEEE Save Cite

The distributed strategy for asynchronous observations in data-driven wildland fire spread prediction

Mengxia Zha Zheng Wang Jie Ji Jiping Zhu Environmental Science, Computer Science ·

International journal of wildland fire · 16 July 2024

TLDR A distributed strategy combined with the Ensemble Transform Kalman filter (ETKF-distributed) for asynchronous observations that can assimilate fire fronts immediately at any location by using new matching schemes between **prediction** and observation is proposed. Expand

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Evaluation of an In-Canopy Wind and Wind Adjustment Factor Model for Wildfire Spread Applications Across Scales

Wei-Ting Hung Patrick C. Campbell +4 authors William Massman Environmental Science ·

Journal of Advances in Modeling Earth Systems · 1 July 2024

The representation of vegetative sub-canopy wind is critical in numerical weather **prediction** (NWP) models for the determination of the air-surface exchange processes of heat, momentum, and trace… Expand

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Learning-based prediction of wildfire spread with real-time rate of spread measurement

Chunjie Zhai Siyu Zhang Zhaolou Cao Xinmeng Wang

Environmental Science, Computer Science · 1 May 2020

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